

HACK NEXUS — Problem Statements

All statements :

Thirteen statements across four domains. Teams pick one.

ID	Title	Domain	Tracks
HN-AI-01	Campus Knowledge Agent	AI & ML	Gemini · MongoDB · Render · Multi-Agent
HN-AI-02	Smart Complaint Triage for Civic Bodies	AI & ML	Gemini · MongoDB · Render
HN-AI-06	Accessibility Auditor for Campus & Civic Websites	AI & ML	Gemini · MongoDB · Render
HN-AI-05	Cross-Hospital Resource Rebalancer	AI & ML	Gemini · MongoDB · Render
HN-CS-01	Phishing Shield	Cybersecurity	Gemini · MongoDB
HN-CS-02	Secure Passwordless Login Kit	Cybersecurity	MongoDB · Render
HN-CS-03	Log Sentinel: AI-Assisted Threat Detection	Cybersecurity	Gemini · MongoDB
HN-CS-04	Tamper-Proof Academic Credential Verification	Cybersecurity	Blockchain · MongoDB · Render
HN-CS-05	Farm-to-Fork Supply Chain Tracker	Cybersecurity	Blockchain · MongoDB
HN-CS-06	Transparent Donation & Fund Tracking	Cybersecurity	Blockchain · MongoDB · Render
HN-CS-07	Decentralized Freelance Escrow	Cybersecurity	Blockchain · MongoDB · Render
HN-FT-01	Smart Expense Companion for Students & Gig Workers	FinTech	Gemini · MongoDB · Render

ID	Title	Domain	Tracks
HN-FT-04	Multi-Agent Portfolio Rebalancer	FinTech	Multi-Agent · MongoDB
HN-FT-06	SME Cash Flow Forecasting	FinTech	Gemini · MongoDB · Render
HN-X-02	Disaster Response Coordination Platform	Cross-Domain	Gemini · MongoDB · Render

A. Artificial Intelligence & Machine Learning

HN-AI-01: Campus Knowledge Agent

Tracks: Gemini · MongoDB · Render · Multi-Agent

Problem: Students waste hours searching for circulars, syllabi, exam timetables and department notices scattered across PDFs, WhatsApp groups and notice boards.

Challenge: Build an AI assistant that ingests institutional documents and answers student queries in natural language. Every answer must cite its source document.

Core deliverable: Upload 5+ PDFs and get a chat interface with grounded answers and source citations. Documents and embeddings are stored in MongoDB.

Stretch goals: Tamil/English bilingual answers, an "I don't know" fallback when the answer is not in the documents, and admin analytics on the most-asked questions.

HN-AI-02: Smart Complaint Triage for Civic Bodies

Tracks: Gemini · MongoDB · Render

Problem: Municipal grievance portals receive thousands of unstructured complaints, often with photos. These are sorted and routed by hand, which causes long delays.

Challenge: Build a system that takes a complaint as text and/or image, classifies its category and urgency, extracts the location, detects duplicates and routes it to the right department.

Core deliverable: A submission form with multimodal classification using Gemini vision and text, an auto-assigned department and priority score, and a dashboard showing complaints by category.

Stretch goals: Merging duplicate complaints, a map heatmap view, and SLA breach alerts.

HN-AI-06: Accessibility Auditor for Campus & Civic Websites:

Tracks: Gemini · MongoDB · Render

Problem: Most college and municipal websites fail basic accessibility standards — poor contrast, missing alt text, unlabeled form fields — locking out visually or motor-impaired users, and manual audits are slow and rarely repeated.

Challenge: Build a tool that scans a given webpage (URL or uploaded HTML) and flags concrete violations with the exact element at fault, plus a ready-to-use fix for each.

Core Deliverable (12h): URL/HTML input, detection for 3 violation categories (alt text, contrast, form labels), a report listing each violation with its element and severity, and a Gemini-generated fix suggestion per violation, logged to MongoDB.

Stretch : A 4th category (heading structure), a before/after preview applying the fixes, a downloadable patch file

HN-AI-05 (revised): Cross-Hospital Resource Rebalance:

Track: Gemini · MongoDB · Render

Problem: Existing bed/oxygen/ventilator dashboards only show each hospital's own stock in isolation — nobody is looking across hospitals in the same district to see that Hospital A is about to run short while Hospital B has spare capacity sitting unused right now.

Challenge: Using a small simulated live feed (5–6 hospitals, one resource type e.g. oxygen cylinders, updated every few seconds) build a system that predicts which hospital will run short within the next few hours using a lightweight trend model, and simultaneously recommends a specific transfer (from which hospital, how much, by when) to prevent it — turning prediction into an actionable redistribution plan, not just a warning.

Core deliverables: Simulated multi-hospital resource stream written to MongoDB → simple trend-based shortage predictor per hospital (e.g., linear depletion rate) with a visible confidence/accuracy check against held-out simulated data → Render dashboard showing live stock per hospital, predicted time-to-shortage, and a ranked transfer recommendation → Gemini-generated one-line justification per recommendation ("Move 40 units from B to A — B has 6hrs surplus, A depletes in 2hrs").

B. Cybersecurity & Blockchain

HN-CS-04 to HN-CS-07 were previously numbered HN-BC-01, HN-BC-02, HN-BC-03 and HN-BC-05. Each still requires a smart contract on a public testnet.

HN-CS-01: Phishing Shield

Tracks: Gemini · MongoDB

Problem: Phishing through SMS, email and fake payment links is the leading cause of digital fraud in India. First-time internet users are the most vulnerable.

Challenge: Build a tool that analyses a suspicious message, URL or screenshot and returns a risk verdict with a simple explanation of why it is dangerous.

Core deliverable: Input as text, URL or screenshot. Output is a risk score plus the red flags found, such as domain age, lookalike domains and urgency language. Every scan is logged to MongoDB.

Stretch goals: A browser extension or WhatsApp-style share target, a community-reported scam database, and regional-language explanations.

HN-CS-02: Secure Passwordless Login Kit

Tracks: MongoDB · Render

Problem: Small web apps built by startups and college teams still rely on weak password-only authentication.

Challenge: Build a drop-in authentication module supporting passwordless login through magic links, OTP or passkeys/WebAuthn, with rate limiting, session management and suspicious-login detection.

Core deliverable: A working demo app using the module, deployed on Render, with at least one passwordless method, brute-force protection and a login activity log.

Stretch goals: Role-based access control, new-device and new-location alerts, and clear documentation so other teams can integrate it.

HN-CS-03: Log Sentinel: AI-Assisted Threat Detection

Tracks: Gemini · MongoDB

Problem: Small organisations generate server logs but have no security team to monitor them, so attacks go unnoticed.

Challenge: Build a system that ingests web server or auth logs, detects anomalies such as brute-force attempts, SQL injection patterns and unusual access, and summarises incidents in plain language.

Core deliverable: Log upload or streaming, rule-based plus ML detection, an incident dashboard, and Gemini-written incident summaries with suggested responses.

Stretch goals: Timeline reconstruction of an attack, email alerts, and mapping to MITRE ATT&CK techniques.

HN-CS-04: Tamper-Proof Academic Credential Verification

Tracks: Blockchain · MongoDB · Render

Problem: Fake degree certificates are a recurring problem, and verifying credentials with universities is slow and manual.

Challenge: Build a DApp where institutions issue certificates whose hashes are stored on-chain. Employers can verify any certificate instantly through a QR code or file upload.

Core deliverable: A smart contract deployed on a testnet such as Sepolia or Polygon Amoy, an institution issuing portal, and a public verification page that flags tampered files.

Stretch goals: Revocation support, bulk issuance, and soulbound tokens for credentials.

HN-CS-05: Farm-to-Fork Supply Chain Tracker

Tracks: Blockchain · MongoDB

Problem: Consumers cannot verify the origin or authenticity of produce labelled organic or locally sourced, and farmers get little credit for quality.

Challenge: Build a supply-chain transparency system where each stage, from farmer to aggregator to transporter to retailer, records a handoff on-chain. Consumers scan a QR code to see the product's journey.

Core deliverable: A smart contract for product batches and handoffs, role-based stakeholder interfaces, and a consumer QR scan page showing the timeline.

Stretch goals: Off-chain storage of metadata and images in MongoDB or IPFS, and a direct tip or payment option to the farmer.

HN-CS-06: Transparent Donation & Fund Tracking

Tracks: Blockchain · MongoDB · Render

Problem: Donors to NGOs, college clubs and relief funds have no visibility into how their money is actually spent.

Challenge: Build a DApp where donations and expenditures are recorded on-chain, funds are released only on milestone approval, and donors can trace their contribution.

Core deliverable: Donations via testnet tokens, milestone-based fund release through a smart contract, and a public transparency dashboard.

Stretch goals: multi-signature approval for fund release, proof-of-spend uploads such as bills and photos, and donor impact reports.

HN-CS-07: Decentralized Freelance Escrow

Tracks: Blockchain · MongoDB · Render

Problem: Freelancers and clients often face disputes over payment, delivery and project completion.

Challenge: Build a smart-contract-based freelance platform where payment is locked in escrow and released according to predefined milestones.

Core deliverable: Project creation, milestone definition, escrow deposit, work submission, approval and payment release.

Stretch goals: A dispute resolution workflow, multi-signature arbitration, and reputation records.

C. FinTech

HN-FT-01: Smart Expense Companion for Students & Gig Workers

Tracks: Gemini · MongoDB · Render

Problem: Students and gig workers have irregular income and limited financial literacy, and they rarely track their spending.

Challenge: Build a personal finance app that auto-categorizes expenses from SMS text, receipts or manual entry, forecasts the month-end balance, and gives personalized savings advice.

Core deliverable: Expense entry including receipt image parsing, auto-categorization, a spending dashboard, and a balance forecast.

Stretch goals: A conversational "Can I afford this?" assistant, savings goals with nudges, and budgeting for irregular income.

HN-FT-04: Multi-Agent Portfolio Rebalancer:

Track: Multi-Agent · MongoDB

Problem: Individual investors can't react to live market shifts fast enough to keep their portfolio aligned with their risk tolerance, and manual rebalancing is slow, error-prone, and reactive rather than continuous.

Challenge: Build a council of collaborating agents — one that scrapes/streams live market data, one that evaluates the user's risk tolerance and target allocation, and one that computes and executes rebalancing trades — coordinating through MongoDB as shared state, with the council converging on a final rebalancing decision.

Core deliverables: Live market data ingestion via web scraping into MongoDB → risk-tolerance agent producing a target allocation profile → rebalancing agent generating trade actions to close the gap between current and target allocation → council/coordinator logic that merges all three agents' outputs into one decision → evaluation dashboard reporting decision time and rebalancing accuracy (drift from target allocation) per run.

HN-FT-06: SME Cash Flow Forecasting

Tracks: Gemini · MongoDB · Render

Problem: Small businesses often make operational decisions without a clear view of their near-term cash position.

Challenge: Build a forecasting system that analyses historical sales, expenses, invoices and payment patterns to estimate future cash flow.

Core deliverable: Dataset ingestion, a cash-flow forecast, an income and expense dashboard, and low-cash alerts.

Stretch goals: What-if scenarios, delayed-payment simulation, and AI-generated business insights.

D. Cross-Domain

HN-X-02: Disaster Response Coordination Platform

Tracks: Gemini · MongoDB · Render

Problem: During emergencies, information about incidents, resources, volunteers and requests for help is fragmented across many communication channels.

Challenge: Build a platform that collects incident reports and intelligently matches requests with available resources and responders.

Core deliverable: Incident reporting, classification, priority assignment, resource matching, and a coordination dashboard.

Stretch goals: Multilingual reporting, duplicate incident detection, map visualization, and AI-generated situation summaries.